CHAPTER III

ORTHOGONAL SETS OF FUNCTIONS

17. Inner Product of Two Vectors. Orthogonality. The concept of an orthogonal set of functions is a natural generalization of that of an orthogonal set of vectors, that is, a set of mutually perpendicular vectors. In fact, a function can be considered as a generalized vector, so that the fundamental properties of the set of functions are suggested by the analogous properties of the set of vectors. In the following discussion of simple vectors, the terminology and notation which apply to the generalized case will be used whenever it seems advantageous for the later generalizations.

Let either g or $g(r)$ denote a vector in ordinary three-dimensional space whose rectangular components are the three numbers $g(1)$, $g(2)$, and $g(3)$. It is the radius vector of the point having these numbers as rectangular cartesian coordinates. The square of the length of this vector, called its *norm*, will be written $N(g)$; it is the sum of the squares of the components of g:

$$N(g) = g^2(1) + g^2(2) + g^2(3) = \sum_{r=1}^{3} g^2(r). \tag{1}$$

If $N(g) = 1$, g is a unit vector, also called a *normed* or a *normalized* vector.

Let θ be the angle between two vectors $g_1(r)$ and $g_2(r)$. Since the components $g_1(1)$, $g_1(2)$, $g_1(3)$ are proportional to the direction cosines of the vector g_1, and similarly for g_2, the formula from analytic geometry for $\cos\theta$ can be written

$$\cos\theta = \frac{g_1(1)g_2(1) + g_1(2)g_2(2) + g_1(3)g_2(3)}{[N(g_1)N(g_2)]^{\frac{1}{2}}}.$$

The numerator on the right is called the *inner product* (or scalar product) of the vectors g_1 and g_2, denoted by the symbol (g_1, g_2); thus

$$\begin{aligned}(g_1, g_2) &= \sum_{r=1}^{3} g_1(r)g_2(r) \\ &= \sqrt{N(g_1)}\,\sqrt{N(g_2)}\,\cos\theta.\end{aligned} \tag{2}$$

When $N(g_2) = 1$, (g_1, g_2) is the projection of the vector g_1 in the direction of g_2.

The condition that the vectors g_1 and g_2 be orthogonal, or perpendicular to each other, can be written

$$(g_1, g_2) = 0 \tag{3}$$

or, in terms of components,

$$\sum_{r=1}^{3} g_1(r)g_2(r) = 0. \tag{4}$$

Note also that expression (1) for the norm of g can be written

$$N(g) = (g, g).$$

18. Orthonormal Sets of Vectors. Given an orthogonal set of three vectors g_n $(n = 1, 2, 3)$, a set of unit vectors φ_n having the same directions can be formed by dividing each component of g_n by the length of g_n. The components of φ_1, for instance, are $\varphi_1(r) = g_1(r)[N(g_1)]^{-\frac{1}{2}}$ $(r = 1, 2, 3)$. This set of mutually perpendicular unit vectors φ_n, obtained by normalizing the mutually perpendicular vectors g_n, is called an *orthonormal set.* Such a set can be described by means of inner products by writing

$$(\varphi_m, \varphi_n) = \delta_{mn} \qquad (m, n = 1, 2, 3), \tag{1}$$

where δ_{mn}, called Kronecker's δ, is 0 or 1 according as m and n are different or equal:

$$\begin{aligned}\delta_{mn} &= 0 \qquad \text{if } m \neq n, \\ &= 1 \qquad \text{if } m = n.\end{aligned}$$

The condition (1) therefore requires that each vector of the set φ_1, φ_2, φ_3 is perpendicular to every other one in that set, and that each has unit length.

The symbol $\{\varphi_n\}$ will be used to denote an orthonormal set whose vectors are φ_1, φ_2, and φ_3. The simplest example of such a set is that consisting of the unit vectors along the three coordinate axes.

Every vector f in the space considered can be expressed as a linear combination of the vectors φ_1, φ_2, and φ_3. That is, three numbers c_1, c_2, c_3 can be found for which

$$f(r) = c_1\varphi_1(r) + c_2\varphi_2(r) + c_3\varphi_3(r) \qquad (r = 1, 2, 3), \tag{2}$$

when the components $f(1)$, $f(2)$, $f(3)$ are given. To find the number c_1 in a simple way, consider equation (2) as a vector equation and take the inner product of both its members by φ_1. This gives

$$(f, \varphi_1) = c_1(\varphi_1, \varphi_1) + c_2(\varphi_2, \varphi_1) + c_3(\varphi_3, \varphi_1) = c_1,$$

since $(\varphi_1, \varphi_1) = 1$ and $(\varphi_2, \varphi_1) = (\varphi_3, \varphi_1) = 0$, according to condition (1). Similarly c_2 and c_3 are found by taking the inner product of the members of equation (2) by φ_2 and φ_3, respectively. The coefficients are therefore

$$c_n = (f, \varphi_n) = \sum_{r=1}^{3} f(r)\varphi_n(r) \qquad (n = 1, 2, 3). \tag{3}$$

The representation (2) can then be written

$$\begin{aligned} f(r) &= (f, \varphi_1)\varphi_1(r) + (f, \varphi_2)\varphi_2(r) + (f, \varphi_3)\varphi_3(r) \\ &= \sum_{n=1}^{3} (f, \varphi_n)\varphi_n(r). \end{aligned} \tag{4}$$

The representation (2) or (4) may be called an "expansion" of the arbitrary vector f in a finite series of the orthonormal reference vectors φ_1, φ_2, and φ_3. These orthogonal reference vectors were assumed to be normalized only as a matter of convenience, in order to obtain the simple formulas (3) for the coefficients in the expansion. The normalization is not necessary, of course.

The definitions and results just given can be extended immediately to vectors in a space of k dimensions. In this case the index r, which indicates the component, has values from 1 to k, instead of 1 to 3; similarly the indices m and n, which distinguish the different vectors of an orthonormal set, run from 1 to k. The definition of the inner product of the vectors g_1 and g_2 in this space, for instance, becomes

$$(g_1, g_2) = \sum_{r=1}^{k} g_1(r)g_2(r). \tag{5}$$

The formal extension to vectors in a space of a countably infinite number of dimensions ($k = \infty$) is also possible. In this case the numbers $g(r)$ ($r = 1, 2, \cdots$) which define a vector g would be so restricted that the infinite series involved, such as the series in (5) with k infinite, would converge. The possibility of the representation corresponding to (2) would have to be examined, of course.

A generalization of another sort is also possible. The units of length on the rectangular coordinate axes, with respect to which the components of vectors are measured, may vary from one axis to another. In such a case the scalar product of two vectors g_1 and g_2 in three-dimensional space has the form

$$(g_1, g_2) = \sum_{r=1}^{3} p(r)g_1(r)g_2(r).$$

The "weight numbers" $p(1)$, $p(2)$, and $p(3)$ here depend upon the units of length used along the three axes.

19. Functions as Vectors. Orthogonality. A vector $g(r)$ in three dimensions was described above by the numbers $g(1)$, $g(2)$, $g(3)$. Any function $g(r)$ which has real values when $r = 1, 2, 3$ will represent a vector if it is agreed that these values are the components of the vector. This function may not be defined for any other values of r, in which case its graph would consist only of three points.

The function $g(r)$ will represent a vector in space of k dimensions if it has real values when $r = 1, 2, \cdots, k$, which are considered as the components of the vector. If $g(r)$ is defined only at these points, it is determined by the vector; graphically it is represented by k points whose abscissas are $r = 1, 2, \cdots, k$, and whose ordinates are the corresponding components of the vector.

Now let $g(x)$ be a function defined for all values of x in an interval $a \leqq x \leqq b$. To consider this function as a vector, the components should consist of all the ordinates of its graph in the interval. The argument x, which has replaced r here, has as many values as there are points in the interval, so that the number of components is not only infinite but uncountable. It is therefore impossible to sum with respect to x as we do with the index r. The natural process now is to sum by integration.

The *norm* of the function or vector $g(x)$, or the sum of the squares of its components, is therefore defined as the number

$$N(g) = \int_a^b [g(x)]^2\, dx. \tag{1}$$

The *inner product* of two functions $g_m(x)$ and $g_n(x)$ is defined as the number

$$(g_m, g_n) = \int_a^b g_m(x) g_n(x)\, dx, \tag{2}$$

in analogy to equation (5), Sec. 18. The condition that the two functions be *orthogonal* is written

$$(g_m, g_n) = 0,$$

or

$$\int_a^b g_m(x) g_n(x)\, dx = 0. \tag{3}$$

Just as before, definition (1) can be written $N(g) = (g, g)$.

A set (or system) of functions $\{g_n(x)\}$ $(n = 1, 2, \cdots)$ is orthogonal in the interval (a, b) if condition (3) is true when $m \neq n$ for all functions of the set. The functions of the set are normed by dividing each function $g_n(x)$ by $[N(g_n)]^{\frac{1}{2}}$, thus forming a set $\{\varphi_n(x)\}$ $(n = 1, 2, \cdots)$, which is normed and orthogonal, or *orthonormal*. An orthonormal system in (a, b) is then characterized as follows:

$$(\varphi_m, \varphi_n) = \delta_{mn} \qquad (m, n = 1, 2, \cdots), \tag{4}$$

where δ_{mn} is Kronecker's δ, defined in Sec. 18. Written in full, equation (4) becomes

$$\begin{aligned} \int_a^b \varphi_m(x)\varphi_n(x)\, dx &= 0 \quad \text{if } m \neq n, \\ &= 1 \quad \text{if } m = n, \ (m, n = 1, 2, \cdots). \end{aligned} \tag{5}$$

The interval (a, b) over which the functions and their inner products are defined is called the *fundamental interval*. Functions for which the integrals representing the inner product and the norm fail to exist must, of course, be excluded.

Throughout this book, only *functions which are bounded and integrable in the fundamental interval, and whose norms are not zero*, will be considered. The aggregate of all such functions for the given interval makes up the *function space* being considered,

in just the same way that the three-dimensional vector space consists of all vectors with three components $g(r)$ $(r = 1, 2, 3)$.

An example of an orthogonal set of functions has already been given in Sec. 15; namely, the functions

$$\sin \frac{n\pi x}{L} \qquad (n = 1, 2, \cdots).$$

The fundamental interval is the interval $(0, L)$. The norm of all these functions is the same number, $L/2$, so the orthonormal set consists of the functions

$$\sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L} \qquad (n = 1, 2, \cdots).$$

The set $\{\sin (n\pi x/L)\}$ is also orthogonal in the interval $(-L, L)$; the normalizing factor is easily seen to be $1/\sqrt{L}$ in this case.

PROBLEMS

1. Show that the set of functions $\{\cos nx\}$ $(n = 0, 1, 2, \cdots)$ is orthogonal in the interval $(0, \pi)$. What is the corresponding orthonormal set? *Ans.* $\{1/\sqrt{\pi}, \sqrt{2/\pi} \cos nx\}$ $(n = 1, 2, \cdots)$.

2. Show that the set $\{\sin x, \sin 2x, \sin 3x, \cdots, 1, \cos x, \cos 2x, \cdots\}$ is orthogonal in the interval $(-\pi, \pi)$. Normalize this set.

20. Generalized Fourier Series. Given a countably infinite orthonormal set of functions $\{\varphi_n(x)\}$ $(n = 1, 2, \cdots)$, it may be possible to represent an arbitrary function in the fundamental interval as a linear combination of the functions $\varphi_n(x)$,

$$f(x) = c_1\varphi_1(x) + c_2\varphi_2(x) + \cdots + c_n\varphi_n(x) + \cdots \qquad (a < x < b). \tag{1}$$

This corresponds to representation (2), Sec. 18, of any vector in terms of the vectors of an orthonormal set.

If the series in equation (1) converges and if, after being multiplied by $\varphi_n(x)$, it can be integrated term by term over the fundamental interval (a, b), the coefficients c_n can be found in the same way as before. Writing the inner product of both members of equation (1) by $\varphi_n(x)$—that is, multiplying (1) by φ_n and integrating over (a, b)—we have

$$\begin{aligned} (f, \varphi_n) &= c_1(\varphi_1, \varphi_n) + c_2(\varphi_2, \varphi_n) + \cdots + c_n(\varphi_n, \varphi_n) + \cdots \\ &= c_n, \end{aligned}$$

since $(\varphi_m, \varphi_n) = \delta_{mn}$. That is, c_n is the projection of the vector f on the unit vector φ_n.

These numbers c_n are called the *Fourier constants* of $f(x)$ corresponding to the orthonormal system $\{\varphi_n(x)\}$; they can be written

$$c_n = \int_a^b f(x)\varphi_n(x)\,dx \qquad (n = 1, 2, \cdots). \tag{2}$$

The series in (1) with these coefficients is called the *generalized Fourier series corresponding to* $f(x)$, written

$$f(x) \sim \sum_1^\infty c_n\varphi_n(x) = \sum_1^\infty \varphi_n(x) \int_a^b f(\xi)\varphi_n(\xi)\,d\xi. \tag{3}$$

The above correspondence between $f(x)$ and its series will not always be an equality. This can be anticipated at once by considering the case of vectors in three dimensions. In that case if only two vectors $\varphi_1(r)$, $\varphi_2(r)$, make up the orthonormal system, any vector not in the plane of those two could not be represented in the form $c_1\varphi_1(r) + c_2\varphi_2(r)$. The reference system here is not complete, in the sense that there is a vector in the three-dimensional space which is perpendicular to both of its vectors φ_1 and φ_2.

Likewise in formula (3), if $f(x)$ is orthogonal to every member $\varphi_n(x)$ of the system, every term in the series on the right is zero, and so the series does not represent $f(x)$.

If there is no function in the space considered which is orthogonal to every $\varphi_n(x)$, the system $\{\varphi_n(x)\}$ is called *complete*. So the system must necessarily be complete if all functions are to be represented by their generalized Fourier series with respect to that system.

PROBLEMS

1. Show that the set $\{\sqrt{2/L}\cos(n\pi x/L)\}$ $(n = 1, 2, \cdots)$ is orthonormal in the interval $(0, L)$, but not complete without the addition of a function corresponding to $n = 0$.

2. Show that the system $\{\sin n\pi x\}$ $(n = 1, 2, \cdots)$ is orthogonal but not complete in the interval $(-1, 1)$.

21. Approximation in the Mean. Let $K_m(x)$ represent a finite linear combination of m functions of an orthonormal set $\{\varphi_n(x)\}$ $(n = 1, 2, \cdots; a \leqq x \leqq b)$; that is,

$$K_m(x) = \gamma_1\varphi_1(x) + \gamma_2\varphi_2(x) + \cdots + \gamma_m\varphi_m(x). \tag{1}$$

The values of the constants γ_n can easily be found for which $K_m(x)$ is the best *approximation in the mean* to any given function $f(x)$; this means the best approximation in the sense that the value of the integral

$$J = \int_a^b [f(x) - K_m(x)]^2\,dx \tag{2}$$

is to be as small as possible; it is also the approximation in the sense of *least squares*.

Writing c_n for the Fourier constants of $f(x)$ with respect to $\varphi_n(x)$,

$$c_n = \int_a^b f(x)\varphi_n(x)\,dx,$$

the integral J can be written

$$\begin{aligned} J &= \int_a^b [f(x) - \gamma_1\varphi_1(x) - \gamma_2\varphi_2(x) - \cdots - \gamma_m\varphi_m(x)]^2\,dx \\ &= \int_a^b [f(x)]^2\,dx + \gamma_1^2 + \gamma_2^2 + \cdots + \gamma_m^2 \\ &\qquad - 2\gamma_1c_1 - 2\gamma_2c_2 - \cdots - 2\gamma_mc_m. \end{aligned}$$

Completing the squares here by adding and subtracting c_1^2, $c_2^2, \cdots, c_m^2$, gives

$$\begin{aligned} J = \int_a^b [f(x)]^2\,dx - c_1^2 - c_2^2 - \cdots - c_m^2 + (\gamma_1 - c_1)^2 \\ + (\gamma_2 - c_2)^2 + \cdots + (\gamma_m - c_m)^2. \end{aligned} \tag{3}$$

It is clear from (2) that $J \geqq 0$, so it follows from equation (3) that J has its least value when $\gamma_1 = c_1$, $\gamma_2 = c_2, \cdots, \gamma_m = c_m$. The result can be stated as follows:

Theorem 1. *The Fourier constants of a function $f(x)$ with respect to the functions $\varphi_1(x), \varphi_2(x), \cdots, \varphi_m(x)$ of an orthonormal set are those coefficients for which a linear combination $K_m(x)$ of these functions is the best approximation in the mean to $f(x)$, in the fundamental interval (a, b).*

Since $J \geqq 0$, it follows from equation (3), by taking $\gamma_n = c_n$, that

$$c_1^2 + c_2^2 + \cdots + c_m^2 \leqq \int_a^b [f(x)]^2\,dx. \tag{4}$$

This is known as *Bessel's inequality*. The number on the right is independent of m; so it follows that the series of squares of the

Fourier constants of any function

$$c_1^2 + c_2^2 + \cdots + c_n^2 + \cdots = \sum_1^\infty c_n^2$$

always converges; and its sum is not greater than the norm of $f(x)$,

$$\sum_1^\infty c_n^2 \leqq \int_a^b [f(x)]^2\, dx. \tag{5}$$

It follows that *the Fourier constants of every function corresponding to any orthonormal system* $\{\varphi_n\}$ *approach zero as* n *tends to infinity:*

$$\lim_{n\to\infty} c_n = 0; \tag{6}$$

because a necessary condition for the convergence of the series in (5) is that its general term c_n^2 approaches zero as n becomes infinite.

22. Closed and Complete Systems. Let $S_m(x)$ be the sum of m terms of the generalized Fourier series corresponding to $f(x)$, with respect to an orthonormal set of functions $\{\varphi_n\}$ $(n = 1, 2, \cdots)$; that is,

$$S_m(x) = \sum_1^m c_n\varphi_n(x). \tag{1}$$

This is the sum $K_m(x)$ in the last section when $\gamma_n = c_n$.

The sum $S_n(x)$ is said to *converge in the mean* to the function $f(x)$ if

$$\lim_{m\to\infty} \int_a^b [f(x) - S_m(x)]^2\, dx = 0. \tag{2}$$

This is also written

$$\underset{m\to\infty}{\text{l.i.m.}}\ S_m(x) = f(x),$$

where the abbreviation l.i.m. stands for *limit in the mean.*

If the relation (2) is true for each $f(x)$ in the function space considered, the system $\{\varphi_n(x)\}$ is said to be *closed.** According to Theorem 1, then, the system is closed if every function can be

* The definitions of the terms closed and complete (Sec. 20) given here are those most commonly used today. Many German writers use the term closed (*abgeschlossen*) to denote what we have called complete, and complete (*vollständig*) for our concept of closed.

approximated arbitrarily closely in the mean by some linear combination of the functions $\varphi_n(x)$.

By expanding the integrand in equation (2) and keeping the definition of c_n in mind, we have

$$\lim_{m\to\infty} \left\{\int_a^b [f(x)]^2\, dx - 2\sum_1^m c_n^2 + \sum_1^m c_n^2\right\} = 0.$$

Hence *for every closed system it is true that*

$$\sum_1^\infty c_n^2 = \int_a^b [f(x)]^2\, dx. \tag{3}$$

This is known as *Parseval's theorem.* When written in the form

$$\sum_1^\infty (f, \varphi_n)^2 = N(f), \tag{4}$$

it identifies the sum of the squares of the components of f, with respect to the reference vectors φ_n, with the norm of f.

Suppose $\theta(x)$ is a function which is orthogonal to every function of the closed set. Substituting it for f in equation (4) gives $N(\theta) = 0$, so that $\theta(x)$ cannot belong to the function space; and thus it is shown that the set is complete (Sec. 20). The following theorem is therefore established:

Theorem 2. *If the set* $\{\varphi_n(x)\}$ *is closed, it is complete.*

It is an immediate consequence that if there is a function which is orthogonal to every member of the set, the set cannot be closed.

This is only a bare introduction to a general theory which has been developed extensively in recent years. To carry it further (even to prove the converse of Theorem 2), a broader class of functions and the idea of the Lebesgue integral are needed.

But the term "closed" was defined here with respect to convergence in the mean, and this type of convergence does not guarantee ordinary convergence at any point. That is, the statement (2) is quite different from the statement of ordinary convergence:

$$\lim_{m\to\infty} S_m(x) = f(x) \qquad (a \leqq x \leqq b).$$

It is this ordinary convergence, and the concept of closed orthog-

that is $[f(x) - S_m(x)]$ can be different

onal sets with respect to it, which are usually needed in the applications.

No general tests of a practical nature exist for showing that a set of functions is closed. That is another reason for deserting the general theory at this point.

23. Other Types of Orthogonality. Some of the important extensions of the concept of orthogonal sets of functions should be noted.

a. A set $\{g_n(x)\}$ $(n = 1, 2, \cdots)$ is orthogonal in an interval (a, b) with respect to a given *weight function* $p(x)$, where it is usually supposed that $p(x) \geqq 0$ in (a, b), if

$$(1) \qquad \int_a^b p(x)g_m(x)g_n(x)\,dx = 0 \quad \text{when } m \neq n \ (m, n = 1, 2, \cdots).$$

The integral on the left represents the inner product (g_m, g_n) with respect to the weight function, a generalization of the inner product of vectors in terms of components with respect to axes along which different units of length are used (Sec. 18). The norm of $g_n(x)$ in this case is, of course,

$$N(g_n) = (g_n, g_n) = \int_a^b p(x)[g_n(x)]^2\,dx \quad (n = 1, 2, \cdots).$$

By multiplying each function g_n of the set by the normalizing factor $[N(g_n)]^{-\frac{1}{2}}$, the corresponding orthonormal set is obtained.

This type of orthogonality can be reduced at once to the ordinary type having the weight function 1. It is only necessary to use the products $\sqrt{p(x)}g_n(x)$ as the functions of the system; then equation (1) shows that the system so formed has ordinary orthogonality in the interval (a, b).

An important instance of orthogonality with respect to weight functions will be seen in the study of Bessel functions later on.

The *Tchebichef polynomials,*

$$(2) \qquad T_n(x) = \frac{1}{2^{n-1}} \cos (n \arccos x), \qquad T_0(x) = 1 \qquad (n = 1, 2, \cdots),$$

also form a set of this type. This set is orthogonal in the interval $(-1, 1)$ with respect to the weight function

$$p(x) = (1 - x^2)^{-\frac{1}{2}}.$$

This is easily verified by integration; thus,

$$\int_{-1}^{1} T_m(x)T_n(x)\frac{dx}{\sqrt{1-x^2}} = \frac{1}{2^{m+n-2}}\int_0^{\pi} \cos m\theta \cos n\theta\,d\theta$$
$$= 0 \qquad \text{if } m \neq n.$$

b. Another extension of orthogonality applies to a system of complex functions of a real variable x $(a \leqq x \leqq b)$. A system consisting of the functions $g_n(x)$, where

$$g_n(x) = u_n(x) + iv_n(x),$$

is said to be orthogonal in the *Hermitian sense* if

$$(3) \qquad \int_a^b g_m(x)\overline{g_n(x)}\,dx = 0 \qquad \text{when } m \neq n,$$

where $\overline{g_n(x)} = u_n(x) - iv_n(x)$, the conjugate of g_n. The system is normed if

$$\int_a^b g_n(x)\overline{g_n(x)}\,dx = 1;$$

that is, if

$$\int_a^b [u_n^2(x) + v_n^2(x)]\,dx = 1$$

for every n.

When the functions are real, $v_n(x) = 0$, and this type reduces to the ordinary orthogonality.

Imaginary exponential functions furnish the most important examples of such systems. For instance, the functions

$$(4) \qquad e^{inx} = \cos nx + i \sin nx \quad (n = 0, \pm 1, \pm 2, \cdots)$$

form a system which is orthogonal on the interval $(-\pi, \pi)$ in the above sense. The proof is left as a problem.

c. Extensions to cases in which the fundamental interval is infinite in length are obtained by replacing a by $-\infty$ or b by ∞, or both.

d. For systems $\{g_n(x, y)\}$ $(n = 1, 2, \cdots)$ of functions of two variables, the fundamental interval is replaced by a region in the xy-plane, and the integrations are carried out over this region. Similar extensions apply when three or more variables are present. Weight functions may be introduced in such cases too, as well as in case *c*.